



Bitcoin: A safe haven asset and a winner amid political and economic uncertainties in the US?

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ABSTRACT

Bitcoin is considered to be an exclusive marvel of the Fourth Industrial Revolution, and is one of the most sophisticated technological and financial products. It has long been a pivot point of attention for investors who are in pursuit of a safe haven asset. In this paper, we use the wavelet-based quantile-on-quantile method, and the quantile-based Granger causality method, in order to investigate the notion of Bitcoin in actually being a safe-haven asset, amid political and economic uncertainty in the US for the period between 2010:M06, and 2020: M10. Using the Partisan Conflict Index (PCI), and the Economic Policy Uncertainty Index (EPU) as proxies of uncertainties, we find that although Bitcoin effectively appears to be a safe haven asset when uncertainties are on the rise, however, this relationship tends to change during the short- to long-run. In this regard, our sample provides us with a unique opportunity to evaluate the safe haven hypothesis for Bitcoin, amid a time span with three Presidential elections in the US, and recently, an ongoing COVID-19 outbreak, which has been declared as a global pandemic. We have also supplemented our analysis with the bootstrap rolling window causality method, as a measure of robustness. In light of the recent COVID-19 pandemic, and the dynamic economic situation, our work provides valuable knowledge for investors, who wish to construct investment portfolios based on Bitcoin, and also provide insights for regulators about how to regulate the cryptocurrency speculation in an effective manner.

1. Introduction

A large number of recent studies have been focused on investigating the possibility of the cryptocurrency market as a new safe haven asset, and a technology-based product (Li et al., 2021; Su et al., 2020d; White et al., 2020). This belief has stemmed from the reasoning that the returns in the traditional asset markets, specifically during periods of increased volatility, have been adversely affected. Thus, it is pertinent to analyze the connection between conventional asset returns, and cryptocurrencies, namely Bitcoin returns. This is a critical endeavor, in order to ascertain how the returns in these market segments behave during turbulent times, while being surrounded by uncertainty. Within the broad range of uncertainties and instabilities that financial markets can experience, the relevant extant literature has mentioned two types of uncertainties that can potentially impact financial markets' returns, including Bitcoin in general. These two uncertainties are political, and

economic in nature.

Political uncertainty also tends to have a significant impact on the performance of financial markets. That is to say that, Political conflicts, rivalries, and factionalism often lead countries towards economic uncertainties, and there is a strongly linked relationship between the two. Political polarization and vagueness may have less predictable, but detrimental effects on the economy through the policy decision-making channels. For instance, while formulating the fiscal policy, political uncertainty and divide can possibly lead markets and economies towards huge economic uncertainty (Azzimonti, 2018; Su et al., 2021b). Therefore, it is a non-trivial task to gauge political uncertainty through the right proxy. A significant amount of literature suggests that, political uncertainty can be measured by examining the divide amongst the political parties, which is also referred to as the partisan conflict. In this regard, political parties represent the collective's interests, and are the main source of collective identity. This collective identity, which

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comprises of the voters and parties, has established within itself a strong relationship (Huddy, 2013). It is noteworthy that these identities are necessary, and indispensable, for the successful functioning of the party process. In recent years, an extensive literature has shown how the party relations tend to promote emotional and political polarization. As the nature of the party identity becomes more common and consistent with the other longstanding party identities (Iyengar et al., 2019; Robison and Moskowitz, 2019), the effects of these processes become overwhelmingly negative. Following this context, many studies have documented how the society judges the social and economic conditions, from a party perspective. Their trust and material advantage, and their relationships are far stronger with partisan identification, than without partisans (McConnell et al., 2018).

In order to investigate the negative consequences of the political uncertainty caused by the partisan conflict, Cheng et al. (2016) investigated the foreign spillover effects of the US partisan, against European countries. Their inspection revealed that the PCI shock would reduce the European industrial production by 0.2%. Decreased production and jobs would, therefore, result in slow exports to Europe, primarily as a consequence of the political shock absorbed by the market. These empirical findings confirm the international instability of the US partisan dispute. In addition to the above, some studies particularly discuss the impact of partisan conflict on financial markets, e.g., by considering the position of the global economic instability, and political disputes. At another instance, Bilgin et al. (2018) investigated the gold pricing determinants, and their findings demonstrated that increasing uncertainty regarding the economic policy would eventually lead to higher gold prices. Besides, it is commonly believed and observed that, when partisan tensions increase, investors perceive gold as a safe haven asset. Moving on, Gupta et al. (2018) took a nonparametric approach towards analyzing the US equity premium forecast, based on the partisan dispute. Their findings showed that partisan conflicts are capable of influencing the stock market volatility, and that partisan conflicts, specifically in the US, can be used as an equity market indicator as well.

Similar to the political uncertainties, it has already been established that the existing literature records a substantial amount of evidence based on the effect of the economic policy uncertainty (EPU) on the financial markets, including stocks (Bakas et al., 2020; Bils et al., 2018; Bouri et al., 2020; Mokni et al., 2020; Qin et al., 2020). Most of these studies suggest that the EPU negatively affects the relationship between these variables. Furthermore, research in this discipline also highlights that the EPU has important portfolio implications that need to be addressed in portfolio management decisions (Badshah et al., 2019). In line with the returns of the equity market, the preliminary literature also investigates the impact of economic uncertainty. This may pertain to, for instance, the market factors, economic policy uncertainty, and the global economic activity (GEA), which are considered to be some of the common macro-economic factors based on the return and volatility estimates in the cryptocurrency market (Gerritsen et al., 2020; Kedar-Levy, 2020; Liu et al., 2019a; Liu and Tsyvinski, 2018; Shaikh, 2020; Umar et al., 2020d). The idea that is being conveyed is that if the cryptocurrency is a safe haven asset, investors will then tend to invest more in the cryptocurrency markets, especially during the times when these uncertainties are high. However, the effect of the EPU on the returns and volatility dynamics that are associated with the Bitcoin, and its subsequent consequences on investment portfolios, is an area of investigation that has not been explored in depth. This is despite the realization that the cryptocurrency market has indicated that the EPU index is an important and significant determinant of the cryptocurrency's price dynamics, particularly for Bitcoin (Bouri et al., 2020; Fang et al., 2019), and can also transmit its associated risks towards the Bitcoin market (Qin et al., 2020). In addition to this, in order to prevent future losses in the Bitcoin market, the EPU can also guide investors in rationalizing their portfolio management decisions (Cheng and Yen, 2020). Nonetheless, the studies looking at the impact of economic policy

uncertainty on Bitcoin market are far less in number, and their results are mostly inconclusive as well.

Taking these intricacies in consideration, the main objective of this paper is to investigate whether Bitcoin indeed emerges as a safe haven asset, particularly when it faces high bouts of political and economic uncertainties in the US. In this regard, this study aims to achieve this goal by considering the impact of political and economic policy uncertainties, on the Bitcoin market's risk and return dynamics, particularly during those time periods that coincide with the US presidential elections. For this purpose, two well-established indices have been taken into consideration, i.e., the PCI and EPU, so as to capture the partisan conflict and economic policy uncertainties, as a proxy of political and economic uncertainties in the US. The sample period taken into account spans from 2010:M07 to 2020:M10, which is a deliberate choice. During this time period, the US economy has recovered from a global financial crisis (GFC), entered into a recent crisis triggered by the COVID-19 situation, and has also experienced three presidential elections. We believe that it is a rare phenomenon to find such a dynamic decade, so as to assess Bitcoin's role as a safe haven asset for investors, with its diversity on both the political and economic fronts. Our results indicate that both PCI and EPU have a positive impact on the bitcoin price (BTC), thus, supporting the argument of bitcoin being a safe haven asset. However, this relationship is not stable and constant, and it tends to move in cycles. That is to point out that, there have been times when both these variables impacted the BTC negatively, like any other financial market. However, the promising aspect here is that, the time and frequency decomposition determined through the wavelet-based quantile-on-quantile analysis, can help us identify whether these cycles exist in the short run, medium run, and/or the long run.

The remainder of this study is arranged as follows: Section 2 presents the literature review. While Section 3 elaborates on the variables description and the empirical modeling that has been used. Moving on, section 4 discusses the empirical results, while section 5 concludes the study by providing the empirical policy implications.

2. Literature review

Bitcoin is a decentralized digital currency, initially proposed as an alternative to fiat currencies. It is independent of the governments of the sovereign, centralized, and banking institutions, and is also considered as an alternative to the inefficient economic and financial standards of the global money markets (Gozgor and Demir, 2018). Bitcoin's appeal to investors and practitioners has been showing a growing trend, especially in times of financial unrest and poor faith, which is one of the reasons that have led to its popularity (Bouri et al., 2017; Luther and Salter, 2017; Shaikh, 2020; Su et al., 2020c). Interestingly though, Bitcoin has become a big subject of interest, in the financial press and academics, as a highly volatile commodity. Given its value and acceptance as an investment tool, Bitcoin's price volatility modeling becomes an indispensable factor, that is closely linked to decision-making and inferences that play a part in the risk management practices (Katsiampa, 2017).

2.1. Role of political uncertainty and partisan conflict in financial markets

Most of the earlier research has a one-way uncertainty effect on the Bitcoin market, or vice versa. BTC and PCI's relationship has hardly been investigated in the extant literature. There is no analysis present that aims to resolve the debate regarding whether Bitcoin is the winner after US factionalism; therefore, the interrelationship between Bitcoin markets, and US factions has not been extensively analyzed. Moreover, the previous studies cannot consider the time-varying parameters of analytical models, and essentially neglect the non-consistent relationship between BTC and PCI. Therefore, this study uses the wavelet-based quantile-on-quantile, quantile granger causality methods, and the robustness check by the rolling window sub-sample bootstrap causality

test, in order to examine the time-changing impact of PCI on BTC. This can essentially address the question as to whether Bitcoin is the winner after US factionalism. Also, by analyzing the unsustainable effect that ripples from BTC to PCI, we can also recognize the Bitcoin market's role in the US political setting.

The impact of partisan rivalry, on corporate cash reserves is explored in detail by (Cheng et al., 2016). Additionally, Hankins et al. proposed three channels to relay the partisan dispute to the relevant cash holdings. It is noteworthy that political division and partisan disputes, in particular, have the capability to reduce investment expenditure as well. Furthermore, at the business level, a decline in spending leads to the access to more cash, especially for the managers. Moreover, the precautionary motive gives the businesses the opportunity to encapsulate more cash holdings in their entirety. If the partisan dispute is usually perceived to be one that extends into the long-term, businesses would often choose to hold their capital, in case of any future political risks. It is significant to point out that a 10% rise in partisan conflict, through a sign restriction strategy, leads to an improvement in the cash holding ratio by 0.4 percent. Moving on from the same context, (Balcilar et al., 2017) used the PCI index to analyze, by employing quantile regression, the causal relationship between party disputes and income inequality. Their empirical findings indicated that there was a decline in the partisan conflict, with the exception of the top quantiles, which tended to decrease the income inequality. Otherwise, because of its power to analyze high-order dependence, causality differed by the quantiles that were taken into consideration.

By considering the specific situation of the global economic instability, and political disputes through a nonlinear ARDL model, Bilgin et al. (2018) investigated, in detail, the gold pricing determinants. Empirical findings demonstrated that the increasing uncertainty about the economic policy would eventually lead to higher prices of gold. The empirical results also showed that when partisan tensions increases, gold is perceived as a safe haven amongst investors. Focusing on the same discipline, Gupta et al. (2018) dwelled into a nonparametric approach, so as to analyze the US equity premium partisan dispute forecast. Their findings showed that partisan conflicts can influence stock market volatility, and that partisan conflicts, particularly in the US, can be used as an equity market indicator as well. Finally, Azzimonti (2018) concentrated on the private investment effect of the partisan dispute. For this purpose, the capital expenditure ratio for late assets was used to describe the investment rates by Azzimonti and Talbert (2014). The empirical findings, using the annual figures, showed a negative correlation between the partisan dispute and the real aggregate investment. In this scenario, it was observed that the dispute would decrease investment rates at the firm level, specifically for the outcomes from quarterly datasets. In short, private investment tend to escalate the increasing partisan dispute. These results support the current hypotheses which pertains to the economic recession due to uncertainties.

2.2. Bitcoin as safe haven during economic uncertainty

Investors have started viewing cryptocurrencies as an attractive investment, owing to the common notion of them being a "safe haven" asset (Dyhrberg, 2016; Wang et al., 2019). The predictors which determine the cryptocurrencies' return, due to the inefficient Bitcoin market, have been investigated by some of the recent studies (Hu et al., 2019). For example, some studies show that crypto-market variables, such as the technical indicators of cryptocurrencies (Gerritsen et al., 2020; Walther et al., 2019), media exposure impacts (Philippas et al., 2020), the return-to-mean property (Turatti et al., 2018) and the conditional tail risks, can be effectively predicted in a timely manner (Bouri et al., 2019a). Moreover, multiple studies show that the returns of cryptocurrencies may also take into account macroeconomic circumstances (Cheng and Yen, 2020). A fewer number of studies analyze whether economic policy uncertainty (EPU) may impact the returns pertaining to cryptocurrencies. Moreover, (Gozgor and Demir, 2018)

also demonstrated that the US EPU index possesses the capability to effectively predict the returns that can be churned out of the Bitcoin. Besides this, Liu et al., (2019) found that Bitcoin and gold could not hedge well against the risk of economic policy uncertainty in the US. In the same context, (Baker et al., 2016) investigated whether the EPU index could predict the returns of cryptocurrencies in different countries. Whereas, the significant impact of US Economic policy uncertainty, on Bitcoin prices, have also been pointed out by (Xia et al., 2020).

A further investigation of the literature shows that, the study conducted by Bouri et al. (2017) concluded that Bitcoin could serve as a hedge, in order to prevent global uncertainty. Bouri et al., (2019) also stressed that Bitcoin can be considered to be a safe haven against any inevitable global financial stress. Furthermore, Gozgor and Demir (2018) revealed that the factor of uncertainty regarding economic policy predicts the Bitcoin's returns, and can be exploited as an asset to put a halt to the level of uncertainty that might be experienced. Bouoiyour et al. (2019) also asserted that Bitcoin could, in fact, act as a hedging instrument, a safe-haven asset, and a portfolio diversifier, in order to cope up with the uncertainty that is experienced in the oil market. Moreover, Bouri and Gupta (2019) pointed out and appreciated Bitcoin's potential to hedge economic uncertainties, and also propagated that these uncertainties can be used to have a more robust prediction about Bitcoin prices in the future. At another instance, Chan et al. (2019) emphasized on Bitcoin's effective hedging role, as the index portfolios consist of five major stock indices. Fang et al. (2019) also proposed that Bitcoin possesses the ability to hedge against the instability that comes attached with the global economic policy, in certain circumstances. In the same stride, a study initiated by (Bermpei et al., 2019) revealed that the correlation between economic uncertainty and the Bitcoin market crashes is negative, and therefore, investors may want to hold on to their Bitcoin reserves, in order to avoid this uncertainty.

Although extensive literature supports the ability of Bitcoin to hedge against uncertainties and risks, this particular belief is not always supported by other researchers who have looked at it, and studied it from another lens. In this regard, Klein et al. (2018) showed evidence of a positive link between Bitcoin and market downturns, essentially suggesting an inconsistency in Bitcoin's hedging abilities. Moreover, Aysan et al. (2019) also stressed upon the revelation that the uncertainty regarding trade policy can significantly affect Bitcoin's returns, and completely eliminate its hedging abilities as well. At another instance, Kliber et al. (2019) asserted upon their findings, and linked Bitcoin's hedging abilities with the exchange location, in which the trade is being made. Matkovskyy and Jalan (2019) also believed that risk-averse investors, specifically during periods of crisis, preferred to steer clear from investing into Bitcoin markets. They examined the EPU's impact on the Bitcoin relationship and the conventional financial markets. Their showed that the volatility spillovers between Bitcoin and conventional markets tend to be higher than the volatility spillovers that occur between conventional markets themselves. In addition to these findings, these volatility dynamics also reveal a complex trend that extends over a period of time, and ends in December 2017, with a frenzy price period, post-launch of Bitcoin's future. The results showed that the EPU tends to have asymmetrical effects on the conventional assets selected. Shahzad et al. (2020) also analyzed how stock market risk investment varies over a period of time. Moreover, (Su et al., 2020d) proposed that Bitcoin cannot always be used as a hedging instrument, thus, suggesting the possibility of its weaker hedging abilities. At another instance, Shahzad et al. (2020) indicated that the efficacy of gold-based hedging tends to be larger than Bitcoin. While, Shaikh (2020) pointed out that the uncertainty of economic policy in the United States and Japan had detrimental effects on the Bitcoin market, while in China, it was beneficial. Finally, stressing upon the discipline of Bitcoin's hedging capacity, Bouri et al., (2019) stressed that it is contingent upon the trade uncertainties. In addition to these studies, Panagiotidis et al. (2020) advocated that EPU is the primary variable that plays a vital role in the prediction and determination of Bitcoin prices, thus depending on a unlikely approach

that is based on the principle of component-led sparse regression. Unlike the extant research and discussion on this topic, by considering the structural shifts in the Bitcoin market, we are able to evaluate the effect of EPU on the complex conditional ties that exist between the Bitcoin and US stock markets, and then determine the portfolio effects, by analyzing EPU's impact on the optimum body weight and hedges.

A careful and meticulous analysis of the literature review helped us in identifying a few gaps and loopholes that we intend to fill through this study. First of all, the impact of political uncertainty on the cryptocurrency market, mainly on Bitcoin, is an area of knowledge that has not been tapped into, or studied till now. Secondly, even though the role of economic uncertainty has been investigated in terms of its authenticity and ability to determine the returns and volatility dynamics of Bitcoin, this initiative has been undertaken in complete isolation, and without any acknowledgement of political uncertainty. Moreover, there is no analysis that has been undertaken, that essentially investigates Bitcoin's role as a safe haven during situations of economic and political uncertainty. In addition to this, the previous studies have not taken into account the time-varying parameters of analytical models, and have completely neglected the nature of the non-constant relationship of Bitcoin prices with economic, or political uncertainty. Therefore, this study uses the wavelet-based quantile-on-quantile, quantile granger causality methods, and the robustness check by the rolling window subsample bootstrap causality test, in order to examine the time-changing effects of uncertainty (both political, as well as economic) on BTC. This will aid in addressing the question as to whether Bitcoin is the winner after US factionalism, which can ascertain whether Bitcoin can be considered as a safe haven or not. Also, by analyzing the unsustainable effect that resonates from BTC to PCI and EPU, we can recognize the Bitcoin market's role in US's uncertain political and economic environment.

3. Variables descriptions and methodology

3.1. Variables descriptions

In order to examine the relationship between Bitcoin price (BTC), partisan conflict (PCI), and economic policy uncertainty (EPU), specifically in the US, this study used the monthly data from 2010M07 to 2020M10. This data constituted of a total of 124 monthly observations, for each variable that has been considered, and came from three independent sources; BTC (Yahoo! Finance¹), PCI (Federal Reserve Bank of Philadelphia²), and EPU (policy uncertainty³). The variables have been used in the natural logarithms, in order to prevent any problems pertaining to heteroscedasticity (Umar et al., 2020c, 2020a, 2020b).

BTC is a digital currency that has been decentralized without a central bank, or a particular administrator. It is readily accessible from one user to another, on a peer-to-peer Bitcoin network that is absent of any intermediaries. Transactions for the BTC are verified cryptographically, using the network nodes. The first Bitcoin exchange, Mt. Gox, was developed in July 2010, promoting and attracting a vast amount of attention to Bitcoin trading. Since shortly afterwards, the advent of BTC has increased dramatically, and Bitcoin is used as an advantage in certain circumstances, in order to escape uncertainty (Qin et al., 2020). To show the Bitcoin market, we therefore chose Bitcoin price (BTC) in USD.

Moving on, the PCI is computed on a monthly basis, with a semantic search approach that is used to measure the frequency of newspaper coverage articles that report political conflict regarding government policies, both within, and among the national parties. In the recent

Table 1

Source of data and descriptive statistics.

	BTC	PCI	EPU
Min.	0.062	34.740	63.877
1st Qu.	24.099	134.820	109.522
Median	478.479	146.934	140.744
Mean	2744.870	150.235	159.293
3rd Qu.	5388.085	162.425	182.352
Max.	15,034.532	271.290	503.963
Standard Deviation	3867.995	73.408	34.257
Skewness	1.236	2.094	0.346
kurtosis	3.182	8.656	6.218
Jarque-Bera	31.728***	255.941***	55.969***
Observations	124	124	124

years, the politics in US has been marked by a high degree of partisan conflict. A higher level of PCI reveals a higher degree of factionalism in the US, and vice versa (Su et al., 2020d).

Lastly, EPU is the economic threat that pertains to the unknown potential policies and government systems. This situation of uncertainty regarding the policies, raises the risk of delayed spending and investment, primarily because of market volatility. Modeling between the BTC and EPU can affect the investors' decision-making ability, specifically when it comes to the distribution of portfolios and hedging strategies (Mokni et al., 2020).

Table 1 provides some of the descriptive statistics of BTC, PCI, and EPU, over the full sample period that has been taken into consideration. In addition to this, contrary to common belief, the Bitcoin prices (BTC) happen to have a kurtosis value that is closer to three, thus refuting the probability of extreme events in the distribution. On the other hand, both the PCI and EPU have a much larger kurtosis value than three, indicating the high likelihood of extreme tail events. In this regard, the Jarque-Bera statistics discard the null hypothesis that propagates that there is normality at a 1% significance level, for all the variables that have been taken into consideration. Besides this, all variables possess a positive skewness, which means that their values are mostly on the rise, over the span of the entire sample period.

The trends pertaining to the BTC, PCI, and EPU are illustrated in Fig. 1. Although the descriptive statistics reflect a positive skewness, all variables mostly seem to be on the rise during the sample period. However, their movement is not completely synchronized with each other, particularly during, and after the US elections that took place in 2016. Although the BTC increased, along with the uncertainties during the US election, however, in 2017, when both the political and economic uncertainties were experiencing a decline, the BTC prices showed an upward spike. Similarly, when in 2019, both uncertainties experienced an upward spike, the BTC declined. These mixed trends demonstrate that BTC has not always proven to be a safe haven for investors, and went through periods of decline, just like the traditional financial instruments behave when uncertainty is on the rise.

In addition to analyzing the original (undecomposed) series, we also

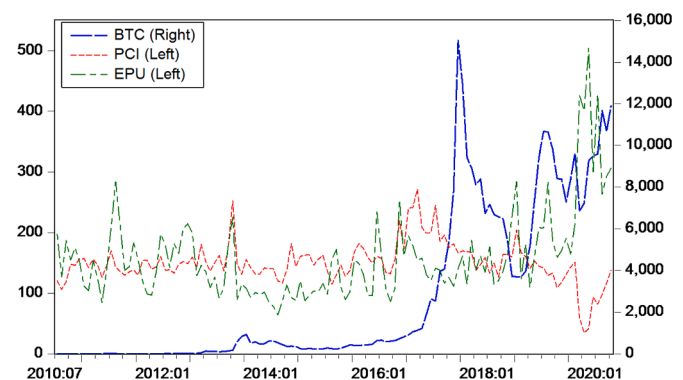


Fig. 1. The trends of BTC, PCI, and EPU.

¹ <https://finance.yahoo.com/quote/BTC-USD/>

² <https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/partisan-conflict-index>

³ https://www.policyuncertainty.com/us_monthly.html

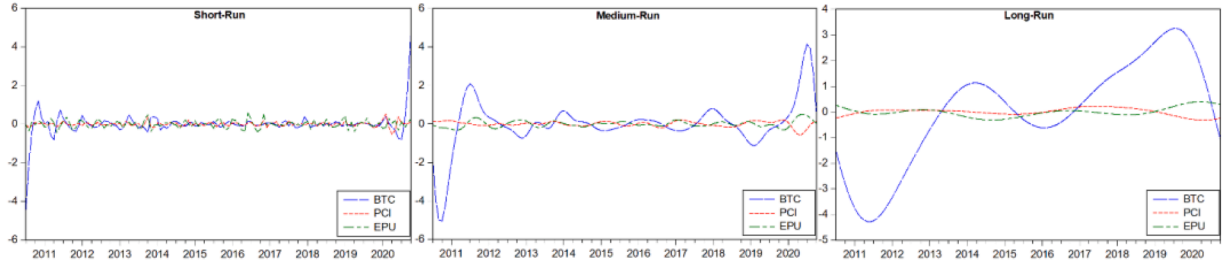


Fig. 2. Series of variables and the wavelet transform.

examined the multiscale wavelet on decomposed series, over the short-, medium-, and long-run trends (Fig. 2). The results showed that the high frequency was set in the short and medium term time. Besides this, the difference in all the other series was very stable for a very long time.

3.2. Methodology

Our methodology adopted in this paper was threefold. The theoretical foundations of our methodology have been based on the Intertemporal Capital Asset Pricing Model (ICAPM), developed by (Cifarelli and Paladino, 2010). This model theoretically explains the transmission mechanism between Bitcoin prices (BTC) and the two proxies of uncertainties, i.e., political uncertainty (PCI) and economic policy uncertainty (EPU). After explaining the transmission mechanism between uncertainty and BTC, we employed a wavelet-based quantile-on-quantile approach and quantile-based granger causality test, in order to analyze the impact of PCI and EPU, on BTC. This approach helped us to decompose the data into the time and frequency domain, and identified the relationship between the variables in the various different quantiles of the decomposed data. Besides there was a strong rationale behind the choice of this approach, that stemmed from the structure of the PCI data. Since the PCI data was available only on a monthly basis, there was an inherent concern that the variations occurring on high-frequency data may be overlooked, and the opportunity to analyze the high-frequency dynamics of the relationship between BTC, PCI, and EPU would be lost, especially if we used other conventional methodologies. Finally, in order to ensure our results' robustness, we employed the bootstrap full sample, and bootstrap rolling window causality tests, so as to analyze the impact of PCI and EPU, on BTC. Further explanation of each set of the methodological approach is provided below.

3.2.1. Theoretical foundations based on ICAPM

The Intertemporal Capital Pricing Asset Model (ICAPM), has been developed by Cifarelli and Paladino (2010). It is particularly helpful in understanding the transmission mechanism between Bitcoin prices (BTC) and macro-level uncertainties (U), which have been proxied by political uncertainty (PCI) and economic uncertainty (EPU). The generalized assumption is that the Bitcoin market tends to have a minimum of two investor(s), and the systematic risk is denoted by (U). In the first instance, this model observes the informed buyers, who take into consideration the dynamics of how risk and returns operate, and who also understand the trends in BTC by analyzing the role that is played by U (CPI and EPU). Thus, keeping these intricacies in consideration, the demand for Bitcoin can be shown in the form of an equation, such as Eq. (1):

$$i_t^D = \frac{E_{t-1}(BTC_t) - BTC^f}{\mu(U_t)} \quad (1)$$

Here, the Bitcoin share, stored by the informed investor, is represented by i_t^D . Moreover, $\mu(U_t) > 0$ and $\mu'(U_t) > 0$ show that all the values are greater than 0, as this is a monotonically increasing function. The price of Bitcoin, without taking into account U, is given by BTC^f .

Moreover, the function, $E_{t-1}(BTC_t)$ denotes the conditional expectation of BTC. While also assuming that the Bitcoin market only caters to informed investors, then $i_t^D = 1$.

By rewriting Eq. (1) $E_{t-1}(BTC_t) = BTC^f + \mu(U_t)$, which was originally developed by (Sharpe, 1964), we can observe that the capital asset pricing model (CAPM) has also been developed. Moreover, it is also clearly known that U exerts a positive effect on BTC. Therefore, the ICAPM can be constructed by considering feedback traders, who take into account the serial correlation of BTC. In this manner, the Bitcoin share stored by the feedback traders is given by $f_t^D = \theta BTC_{t-1}$, where $\theta > 0$. Assuming that the Bitcoin market has two kinds of investors, we can have the function $i_t^D = 1 - f_t^D$, which can then be formulated into Eq. (2), by transforming Eq. (1):

$$E_{t-1}(BTC_t) = BTC^f + \mu(U_t) - \theta \mu(U_t) BTC_{t-1} \quad (2)$$

Here we can observe that the coefficient of $\mu(U_t)$ is $1 - \theta BTC_{t-1}$, and it is a positive value since $\theta BTC_{t-1} = f_t^D < 1$. Therefore, the results may show that BTC may be affected positively by U (PCI and EPU). Moreover, a high U may lead BTC to increase, so as to compensate for the losses incurred due to political and economic uncertainty. These assumptions are completely in line with the trend that we observed in Fig. 1, when in 2016, political uncertainty and PCI rose due to the presidential election campaign, and the conflicts between the Democrats and the Republicans became increasingly fierce and competitive. At that time, investors were inclined towards storing hedge-able assets (for example, Bitcoin), and the corresponding behavior and search for a safe haven asset eventually pushed up the price of BTC, that too significantly.

3.2.2. Wavelet-Based quantile on quantile approach

Since a long time, researchers have begun to realize the importance, and comparative superiority of the Quantile Regression Analysis (QRA), that has originated from the teachings of (Koenker and Bassett Jr, 1978), over linear regression. However, despite its ability to estimate and analyze the heterogeneous relationship at different quantiles of the conditional distribution, the QRA is unable to accommodate the possibility of varying relationships that are caused by the nature of the uncertainty. In order to deal with this issue particularly, Sim and Zhou (2015) proposed the novel approach of Quantile on Quantile (QQ). Therefore, we modeled the relationship between BTC and the two types of uncertainties that pertain to PCI and EPU, using the QQ approach in this paper. This approach is capable of providing a comprehensive picture of the nature of BTC's relationships with uncertainty, at different quantiles of the conditional distribution.

Thus, our nonparametric model, based on the quantile regression would look like the following function;

$$BTC_t = \beta^\theta U_{t-1} + u_t^\theta \quad (3)$$

where, BTC_t represents the Price of Bitcoin, during time period t; and U_t can take either of the two forms that represent the Political Uncertainty, captured through (PCI), or economic uncertainty captured through (EPU), at day t. Moreover, θ is the θ th quantile of the conditional distribution of BTC_t , and u_t^θ is a quantile error term, whose conditional θ th quantile is equal to zero. Since $\beta^\theta(\cdot)$ is an unknown function, primarily because there was no prior information linking BTC_t and U_t , we, therefore, estimated the following equation, using the first-ordered

Taylor expansion;

$$\beta^\theta(U_{t-1}) = \beta^\theta(U) + \beta^{\theta'}(U)(U_{t-1} - U^\tau) \quad (4)$$

Rewriting Eq. (5) as (6), and then substituting (6) in (4) would yield us Eq. (7);

$$\beta^\theta(U_{t-1}) = \beta_0(\theta, \tau) + \beta_1(\theta, \tau)(U_{t-1} - U^\tau) \quad (5)$$

$$BTC_t = \underbrace{\beta_0(\theta, \tau) + \beta_1(\theta, \tau)(U_{t-1} - U^\tau)}_{(*)} + u_t^\theta \quad (6)$$

Additionally, estimating Eq. (7) requires replacing U_{t-1} and U^τ with their estimated counterparts $\hat{U}_{t-1}$ and $\hat{U}^\tau$, respectively. The local linear regression estimates of the parameters b_0 and b_1 , which are the estimates of β_0 and β_1 , are therefore obtained by solving the following minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^n \rho_\theta \left[BTC_i - b_0 - b_1 \left(\hat{U}_{i-1} - \hat{U}^\tau \right) \right] \times K \left(\frac{F_n(\hat{U}^\tau) - \tau}{h} \right) \quad (7)$$

Where $\rho_\theta(u)$ is the quantile loss function, defined as $\rho_\theta(u) = u(\theta - I(u < 0))$, and I being the usual indicator function. Additionally, $K(\cdot)$ denotes the Gaussian kernel function, and h is the bandwidth parameter of the kernel. Following the teaching of (Sim and Zhou, 2015), we chose $h = 0.05$ as the bandwidth parameter, in this paper.

In order to further enrich the analysis, we combined the QQ approach with the multiscale wavelet decomposition. In this context, the Wavelet analysis refers to a combination of the time and frequency domain, of a variable's time-series data. The major advantage of the Wavelet analysis is that, we can get the nonparametric representation of individual time series by decomposing the data into an orthogonal timescale, as put forth by Ramsey (1999). This decomposition in the Wavelet scales keeps the time data preserved, and allows us to effectively deal with the non-stationary part of time series data.

The Wavelet also requires the orthonormal bases that have been obtained by dynamically dilating, and translating a pair of specially constructed functions φ and, such that:

$$\int_{-\infty}^{\infty} \varphi(t) dt = 1 \quad (8)$$

$$\int_{-\infty}^{\infty} \psi(t) dt = 0 \quad (9)$$

Where φ and ψ are referred to as the father wavelet and mother wavelet, which can be used to capture the low-frequency (smooth), and high-frequency (detailed) parts of the series, respectively.

Then the obtained Wavelet basis can be represented, respectively, by the following pair of functions:

$$\varphi_{j,k}(t) = 2^{j/2} \varphi(2^j t - k) \quad (10)$$

$$\psi_{j,k}(t) = 2^{j/2} \psi(2^j t - k) \quad (11)$$

Where $j = 1, \dots, J$ denotes the scale, and $k = 1, \dots, 2^j$ denotes the subsequent translation. It is noteworthy that the maximum number of scales that can be considered in the analysis happens to be limited by the number of observations ($T \geq 2^J$) that are taken into account.

3.2.3. Quantile-Based granger causality test

We further supplemented our Wavelet based Quantile on Quantile analysis by incorporating the quantile based granger causality. Based on Granger (1969), if a series Z_t does not contribute towards predicting another series Y_t , then it is said that Z_t does not granger-cause Y_t . Mathematically, let there be an explanatory vector $I_{t|del} (I_t^Y, I_t^Z)' \in \mathbb{R}^d$, where $d = s + q$, and I_t^Z , which is the past information set of $Z_t, I_t^Z := (Z_{t-1}, \dots, Z_{t-q})' \in \mathbb{R}^q$. The mathematical equation for the null hypothesis of granger non-causality from Z_t to Y_t is therefore given below:

$$H_0 : F_Y(y | I_t^Y, I_t^Z) = F_Y(y | I_t^Y), \text{ for all } y \in \mathbb{R} \quad (12)$$

Here, the function $F_Y(\cdot | I_t^Y, I_t^Z)$ represents the conditional distribution of Y_t , given (I_t^Y, I_t^Z) . Since the conditional distribution estimates may be complicated in nature, literature has proposed tests for granger causality in mean (Troster et al., 2018). In this case, Z_t does not granger-cause Y_t in the mean if,

$$E(Y_t | I_t^Y, I_t^Z) = E(Y_t | I_t^Y), \text{ a.s.,} \quad (13)$$

However, in order to analyze the Granger non-causality in conditional quantiles, we defined $Q_T^{Y,Z}(\cdot | I_t^Y, I_t^Z)$ as the τ -quantile of $F_Y(\cdot | I_t^Y, I_t^Z)$. Thus, we rewrite the null hypothesis as given below, where T refers to the compact set such that $T \subset [0, 1]$.

$$H_0 : Q_T^{Y,Z}(Y_t | I_t^Y, I_t^Z) = Q_T^Y(Y_t | I_t^Y), \text{ a.s. for all } \tau \in T \quad (14)$$

For further details on the methodology, interested readers can pursue (Troster et al., 2018).

3.2.4. A robustness measure

3.2.4.1. Bootstrap full-sample causality test. As a final point, to guarantee the uniformity of our results, we employed the Residual Bootstrap Method (RB) that has been developed by (Shukur and Mantalos, 1997), in order to confirm the robustness of our results from the Quantile on Quantile, and the quantile based granger causality methods. Shukur and Mantalos (1997) stressed upon the belief that their approach can be deemed to be suitable and appropriate, even when the time series is limited enough for statistics to follow the standard asymptotic distributions. They also proposed the measure of the probability ratio (LR), which has been updated using the power and size characteristics. We, therefore, analyzed the relationship of BTC with PCI and EPU, using the RB-based modified LR statistics. In addition to this, since fluctuations in PCI can simultaneously affect the BTC (Su et al., 2020d), and EPU (Dyhrberg, 2016; Baur et al., 2018; Qin et al., 2020), the correlation in one pair of variables can be affected by the third variable as well (Fang et al., 2019; Gozgor and Demir, 2018; Su et al., 2020b). Therefore we allowed all the three variables to simultaneously interact with each other in the VAR model, as presented in Eq. (15):

$$\begin{bmatrix} BTC_t \\ PCI_t \\ EPU_t \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \gamma_{20} \\ \gamma_{30} \end{bmatrix} + \begin{bmatrix} \gamma_{11}(L) & \gamma_{12}(L) & \gamma_{13}(L) \\ \gamma_{21}(L) & \gamma_{22}(L) & \gamma_{23}(L) \\ \gamma_{31}(L) & \gamma_{32}(L) & \gamma_{33}(L) \end{bmatrix} \begin{bmatrix} BTC_t \\ PCI_t \\ EPU_t \end{bmatrix} + \begin{bmatrix} \mu_{1t} \\ \mu_{2t} \\ \mu_{3t} \end{bmatrix} \quad (15)$$

Here, the white-noise process is given by $\mu_t = (\mu_{1t}, \mu_{2t})' \cdot \gamma_{ij}(L) = \sum_{k=1}^p \gamma_{ijk} L^k$, where $i = 1, 2; j = 1, 2, 3, 4; p$ denotes an optimal lag order; L shows the presence of a lag operator, and there is $L^k Z_t = Z_{t-k}$. The null hypothesis that propagates that the PCI or EPU cannot affect BTC, that is $\gamma_{12,k} = 0$, and $\gamma_{13,k} = 0$, can therefore be rejected when PCI or EPU do Granger cause BTC, and vice versa. Likewise, the null hypothesis, that proposes that BTC has no effect on PCI ($\gamma_{21,k} = 0$) or EPU ($\gamma_{31,k} = 0$), can also be examined in this manner.

3.2.4.2. Parameter stability test. The above test assumes that the VAR system parameters are constant and not practically or empirically compliant. Therefore, if the parameters are unstable, they may not be applicable to the full sample test. In order to verify whether the parameters obey the random walk processes, we conducted parameter stability tests, containing the Sup-F, Ave-F, and Exp-F tests (Hansen, 1992; Nyblom, 1989), that are capable of assessing the structural changes, as well as the Lc statistics (Nyblom, 1989; Hansen, 1992). The non-stable link between BTC, PCI and EPU can thus be calculated by these four tests, if there is a presence of an unstable parameter. The sub-sample test should also be used to evaluate the association between BTC and PCI, and BTC and EPU.

3.2.4.3. Bootstrap sub-sample rolling-window causality test. In order to divide the total sample into smaller parts, based on the rolling window width w , the method pertaining to this particular subsample was developed by Balcilar et al. (2010). Each of the section finals have w , $w + 1$, ..., S , and $S - w + 1$ sub-samples, that can be obtained. Moreover, each of these sub-samples can also achieve a valid result by conducting the RB-based updated LR statistics, from the Granger causality test. Thus, the sub-sample test results can be obtained by taking into consideration the following functions: $N_b^{-1} \sum_{k=1}^p \hat{\gamma}_{12,k}^*$ and $N_b^{-1} \sum_{k=1}^p \hat{\gamma}_{21,k}^*$, which are the averages of large estimations. These essentially show the impact from PCI and EPU, to BTC and the effect of BTC to PCI, and EPU, respectively. Moreover, the times of the bootstrap iterations are represented by N_b , followed by $\hat{\gamma}_{12,k}^*$, $\hat{\gamma}_{13,k}^*$ and $\hat{\gamma}_{21,k}^*$, $\hat{\gamma}_{31,k}^*$, which are the estimations that have been made through Eq. (3). This paper applies a 90% confidence interval, and the related lower (5th quantile) and upper (95th quantile) limits (Su et al., 2020e); (Su et al., 2021a); (Su et al., 2020a).

4. Empirical findings and discussions

This section presents the preliminary empirical findings of the QQ analysis, by using both the original series, and the series that proceeds the Wavelet decomposition. In this regard, Fig. 3 displays the impact of PCI on BTC, and the impact of BTC on PCI, specifically in the context of the US market. The values observed on the z-axis capture the effect of the n th quantile of PCI (BTC), on the n th quantile of BTC (PCI), at varying scaling coefficients. Fig. 3 also reveals some important results, and shows that the relationship between BTC and PCI does not remain stationary or stagnant during the time period that transitions from the short run to the long run.

Furthermore, results of the original series (Fig. 3, column 1) are indicative that the PCI (in different quantiles) on BTC (in different quantiles) exhibit differences. Thus, PCI's strong positive effect on BTC is found in the medium to upper quantiles of PCI (0.40–0.95), with the lower quantiles of BTC (0.05–0.10). On the other hand, a negative effect of PCI on BTC is visible, at the lowest quantiles of PCI (0.05–0.40), and the lower to highest quantiles of BTC (0.20–0.95); but this relationship exhibits a weaker link. To further explore the short to long-run transitional relationships between BTC and PCI, we have used the Wavelet transformation, in order to decompose the time series of variables into three different frequencies; including the short, medium and long-run frequencies. In this context, column 1 of Fig. 3, and rows 2–4 display these decomposed relationships between BTC and PCI. It is evident that during the short run, PCI's positive effect on BTC is clearly visible in the medium to upper quantiles of both PCI and BTC (0.40–0.95). On the other hand, a negative effect of PCI, on BTC is found within the lowest quantiles of PCI (0.15–0.30), and the lower quantiles of BTC (0.10–0.30). During the medium-run, the effect of PCI on BTC tends to be positive. This effect is only robust at the middle quantiles of PCI (0.60–0.70) and BTC (0.60–0.80), and at the PCI and BTC peak quantiles. When considering the long-run, it is observed that PCI's effect on BTC is strongly positive in the area combining the lower to upper quantiles of BTC (0.2–0.95), and the intermediate quantiles of PCI (0.20–0.40). A weak negative effect of PCI on BTC is also observed for medium and long-term estimates. Overall, the findings show that the effect of PCI on BTC, in different quantiles, tends to have a positive and negative relationship.

Results of the original series (Fig. 3, column 2) also show the impact of BTC (in different quantiles) on PCI (in different quantiles). The association between BTC and PCI is positive, from moderate to strong in the medium to upper quantiles of PCI (0.50–0.95), and the medium quantile in BTC (0.40–0.60). The weakest negative relationship has been observed in the remaining quantiles. Columns 2 of Fig. 3, and rows 2–4 refer to the relationship between BTC and PCI, in the short to the long-run. In the short-run, BTC's effect on PCI is negative and weak in all the quantiles of BTC and PCI, while a relatively high effect is found in areas combining the high quantiles of PCI, and the medium quantile of BTC. In

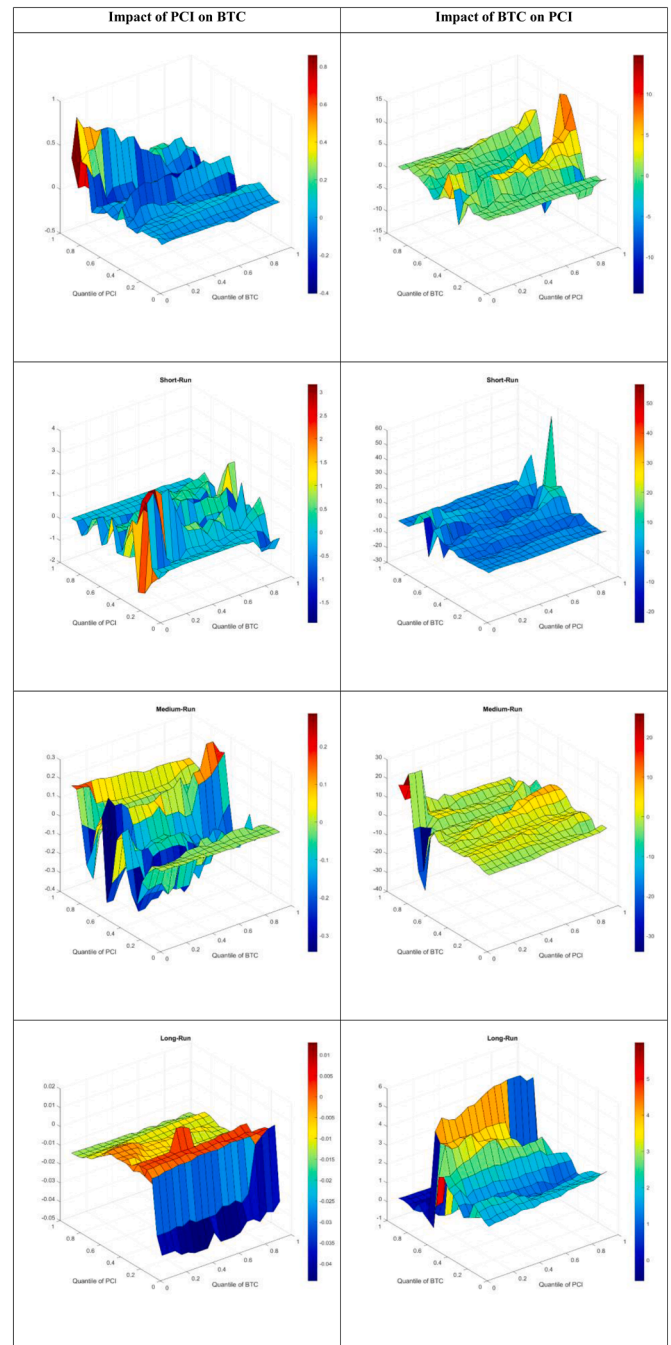


Fig. 3. Q-Q estimates of the slope coefficients (BTC and PCI).

the medium- to long-run, there is a positive and negative association between BTC and PCI. The positive effect of BTC on PCI is visible only at the middle to upper quantiles of BTC (0.75–0.95), and the lower quantiles of PCI (0.10–0.20). In contrast, BTC's negative effect on PCI is found at the lower to middle quantiles of both BTC and PCI. Overall, the findings show that the effect of BTC on PCI, in different quantiles, is also exhibiting a mixed trend.

The position of Bitcoin, as a safe haven, and the notion that it may be a winner during factionalism and high bouts of political uncertainty in the US, can therefore be supported by the positive effects that PCI exhibits on BTC. There are multiple factors that influence this development. Firstly, after the 2016 presidential election, nearly two-thirds of the American population observed that there was an intense conflict between the Democrats and Republicans in the US. The Trump

administration instituted new policies such as removing Obamacare, withdrawing from the Trans-Pacific Partnership TPP Agreement, and building walls on the US-Mexico border, which increased the conflict between the two parties. The surge in PCI, therefore led to a decline in investor sentiment and consumer trust, pushing them to hold assets with hedging potential (such as Bitcoin), which further increased the demand for Bitcoin (BTC), and also its price subsequently (Jiang et al., 2020). In addition to these developments, the decisions taken by the Trump administration resulted in a high degree of ambiguity in the general society (Pham et al., 2018), and thus, the investors reduced their US dollar holdings, and stored in more Bitcoin, which further increased the BTC.

Another deciding factor would be the recent COVID-19 outbreak and its spread to the planet. During this time, the Trump administration's actions to combat the epidemic proved to be highly ineffective. With the continuous spread of COVID-19 in the US, the political parties' discontent increased, and the PCI thus expanded. However, the pessimistic feelings of investors, due to the delay in the arrival of COVID-19 vaccination, could not stop them from investing in Bitcoin, and therefore, BTC has been growing significantly during this outbreak as well. Last but not least, it has also been observed that during the final period of the election campaign of 2020, the high degree of political conflict between President Donald Trump, and his Democratic challenger Joe Biden, was reflected by the upward spike in the PCI, which has further pushed the prices of BTC upward.

However, on the other hand, we need to keep in mind that the negative effect of the PCI on BTC does not empirically support the argument that Bitcoin is a safe haven - particularly during political uncertainty. After an unprecedented boom of \$14,000 in 2017, Bitcoin's price fell by approximately 65% in the months spanning from January to February 2018. In this case, firstly, Democrats and Republicans conflicted when President Trump tweeted that his "Nuclear Button" is larger, and more effective than North Korea's arsenal. It was recorded that the Democrats quickly condemned Trump's tweets, and called on him to immediately give up on further increasing the tensions with the rogue country. Secondly, the US dollars partially appreciated the US economy's development (i.e., imposed tariffs on goods imported from China), and a raised the interest rate for the Federal Reserve Board. As a result, Trump resorted to the imposition of tariffs on US\$50 billion worth of Chinese imports, and by the time it was July 2018, he and his administration raised the tariff amount to US\$250 million, claiming that the trade war is a good initiative that is easy to get over. However, this trade war had significant adverse effects on the US economy, and failed to remove any economic hindrances for the US as well. In fact, by the end of the year, these policies made by Trump actually hurt the American economy, instead of helping to elevate or even restore it to some extent. Furthermore, the stock market was also sold off, which led to a wider trade deficit for the US, and as a result, trading partners to the US ended up imposing retaliatory tariffs on US exports. This then caused the US farmers to lose their access to overseas markets, and also led to a massive drive of laying off employees by some prominent US manufacturers. The primary reason for this was that the higher input costs hurt the bottom line of the operating companies. In this regard, the US Senate has hereon restricted Trump's tariff power by an overwhelming majority. Without a doubt, these restrictive measures, and the conflict with China have increased the PCI, but at the same time, BTC has gradually experienced a decline, as the US dollars have appreciated (Baur et al., 2018). Moreover, there are also certain other explanations for the negative relationship between PCI and BTC. These include factors such as the worse than expected performance of Bakkt's futures (a digital assets custody service), that has reduced the investor confidence in the digital currency market, and at the same time, pushed the BTC prices downward. Besides this, the US Securities and Exchange Commission (SEC) has also postponed its ruling on the three cryptocurrency exchange-traded funds (ETFs). This change in the regulators' stance has made the Bitcoin market's positive prospects worse than expected,

therefore resulting in a decline in the Bitcoin holdings and BTC.

After examining the political uncertainties (PCI), it is also important to examine the impact of economic policy uncertainty (EPU) on BTC, and vice versa. In this regard, Fig. 4 shows the impact of the US EPU on BTC, and the impact of BTC on EPU. The value on the z-axis captures the effect of the nth quantile of EPU (BTC), on the nth quantile of BTC (EPU), at varying scales. Fig. 4 also reveals critical results that reflect that the relationship between BTC and EPU is, in fact, not consistent and stable when transitioning from the short to the long-run. Moreover, the results of the original series (Fig. 4, column 1) show that the impact of EPU (in different quantiles) on BTC (in different quantiles), is therefore, also varying. Additionally, for higher quantiles of BTC and EPU, the impact of EPU on BTC tends to be positive. However, in the lower quantile of EPU, and the lower to high-quantile of BTC, EPU has a negative impact

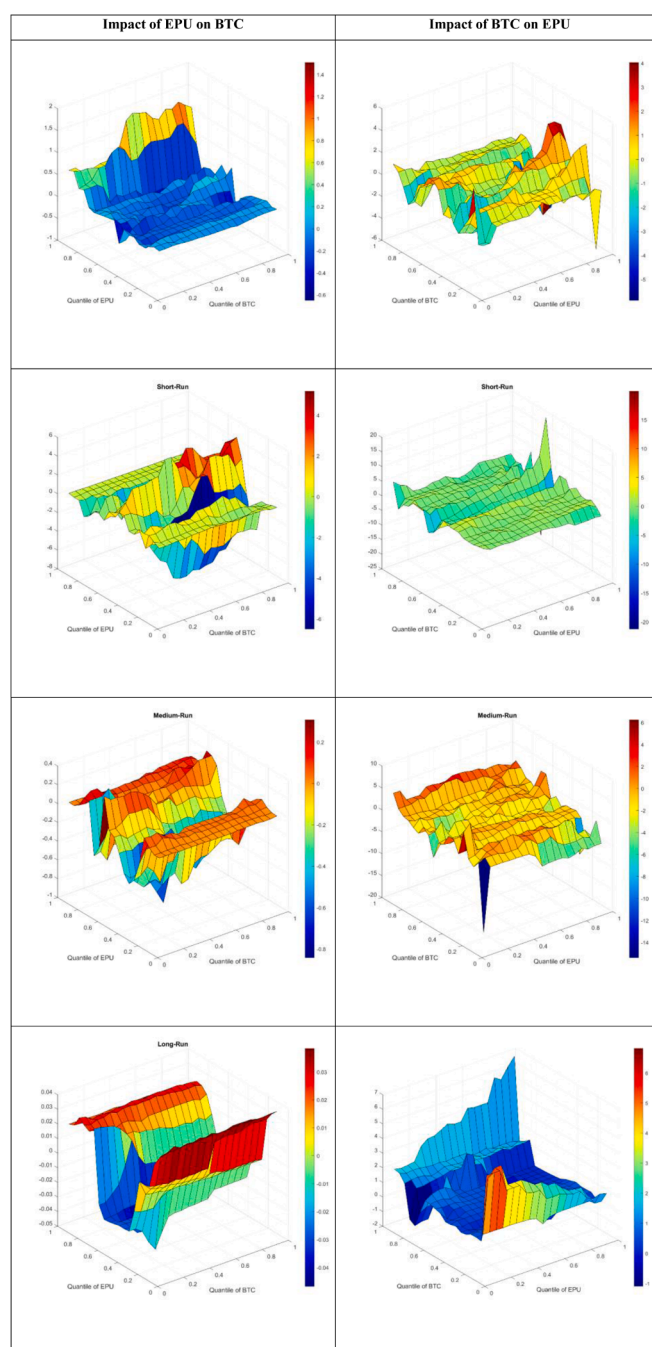


Fig. 4. Q-Q estimates of the slope coefficients (BTC and EPU).

on BTC. Similar to the PCI, to know the differences in the transition from the short to long-run relationships, the Wavelet transformation has been referred to. Column 1 and rows 2–4 of Fig. 4 show that in the short run, the impact of EPU on BTC is minimal, and is only positive within the low to the medium quantile of EPU (0.35–0.50), and the middle to high quantile of BTC (0.60–0.90). However, this effect becomes very strong in the medium run and long run for the lower quantile of EPU, and the low to the high quantile of BTC. Not much different from the findings of PCI, these results also support the mixed effect of EPU on BTC, in different quantiles.

Moving on, column 2 of Fig. 4 displays the impact of BTC (in different quantiles), on EPU (in different quantiles). This association between BTC and EPU is positive in the medium to upper quantiles of EPU (0.65–0.95), and in the medium quantile in BTC (0.50–0.60). For the remaining quantiles, the weakest negative relationship has been observed. In addition to this, column 2 of Fig. 3, and rows 2–4 show that during the short-run, BTC's effect on EPU is mixed in nature, and is also weak in all the related quantiles of BTC and EPU. In the medium -run, there is a positive and negative association between BTC and EPU. The positive effect of BTC on EPU only seems to be visible at higher quantiles. In contrast, BTC's negative influence on EPU is found at the low to middle quantiles, of both BTC and EPU. This strong positive relationship is found at a low quantile of both EPU and BTC in the long-run.

Overall, the findings show that the effects of EPU on BTC, in different quantiles, have a positive and negative relationship. That is to say that, the positive impact of EPU on BTC suggests that Bitcoin may indeed be considered as a safe haven asset. However, the increase in policy uncertainty certainly makes investors pessimistic about the international economic environment, and compels them to invest in profitable, and safe assets. If Bitcoin is really that asset, the increased demand of Bitcoin, and subsequently the increased prices can offset the losses caused by the increased EPU. When looking at examples of other countries, we can consider that after Argentina's exchange rate and Turkey's debt crisis, EPU rose in 2018, and so did the BTC. Similarly, from December 2018 to January 2019 (35 days), the US government imposed a shutdown as well. This shutdown not only led to the intensification of the partisan conflicts in the US, but also increased the EPU significantly. However, with the end of the crisis, and the government's shutdown, China and US agreed not to impose any new tariffs, and thus, the EPU declined, and so did the BTC. The reason of this decline in BTC is straightforward, as the decline in EPU strengthened the public's trust in the world economy, and also revived the investors' confidence in financial markets. As a result, they reduced their demand for safe-haven assets (such as Bitcoin), and reallocated their funds towards high yielding assets that then caused a decrease in BTC.

After estimating the variables, this study uses the quantile based Granger causality method, so as to examine the causality between the selected variables. This method allows us to evaluate data for causality measures that are tested on different quantiles. In this regard, tables 2 and 3 summarize the required lag order, and the joint significance test results. The test statistics values overwhelmingly reject the 5% non-causal relationship, in the low quantile range (0.05–0.95) and (0.05–0.20), which shows that PCI can indeed affect BTC, and BTC on PCI only leads to the highest quantile range (0.80–0.95).

As for all the other time intervals that have been taken into

Table 2
Quantile causality test: BTC and PCI.

Quantile interval	lag Order	PCI $\neq$ BTC	lag Order	BTC $\neq$ PCI
[0.05~0.95]	1	9.460**	1	2.897
[0.05~0.20]	1	9.460**	1	2.733
[0.20~0.40]	1	0.997	1	1.518
[0.40~0.60]	1	0.601	1	0.352
[0.60~0.80]	1	1.241	1	0.202
[0.80~0.95]	1	7.150**	4	25.480***

Note: *** and ** denote significance at 1% and 5% level.

Table 3
Quantile causality test: BTC and EPU.

Quantile interval	lag Order	EPU $\neq$ BTC	lag Order	BTC $\neq$ EPU
[0.05~0.95]	2	11.803**	1	2.872
[0.05~0.20]	2	11.803**	1	1.194
[0.20~0.40]	1	2.134	1	1.213
[0.40~0.60]	1	0.469	1	2.556
[0.60~0.80]	1	1.976	1	2.896**
[0.80~0.95]	1	1.496	1	0.608

Note: ** denote significance at 5% level.

consideration, the test statistics do not seem to be significant. Moreover, there have been observations of a two-way dependence between the EPU and the BTC, that too at a low quantile interval (0.05 – 0.95) and (0.05 – 0.20) on BTC, while the medium quantile interval (0.60 – 0.80) for the BTC on EPU. Furthermore, it has also been observed that the test statistics of the quantile intervals are significant in nature, when considering the effects of PCI and EPU on BTC. However though, this is specifically applicable when the latter are significantly high or low. When taking into account the robustness checks, this study uses the bootstrap rolling window causality method, in order to confirm the findings of the quantile-on-quantile, and quantile based Granger causality methods. Firstly, this study used the VAR method, based on Eq. (15), to measure all the samples of causality characteristics between the selected variables. In this regard, Table 4 reports the results of the full sample bootstrap Granger causality test. As observed, no apparent correlation was found between the variables that have been taken into account, which indicates that the PCI and EPU do not cause the fluctuations in BTC, and vice versa. However, these findings contradict the existing literature on this discipline (Bouri et al., 2017; Bouoiyour et al., 2019; Bouri and Gupta, 2019), and the ICAPM hypothesis emphasizes that PCI and EPU may in fact be beneficial to BTC.

In Eq. (15), the full sample's estimation assumes that no parameters are linked with time, and there is only one Granger causality in the sample. However, when there are any structural changes in the variables and the VAR model, the causal link between variables can exhibit the time-variation characteristic (Balcilar and Ozdemir, 2013; Qin et al., 2020). From here on, following Andrews (1993), and Andrews and Ploberger (1994), the *Sup-F*, *Mean-F*, and *Exp-F* tests have been applied in this study, so as to examine the stability parameters. Besides this, the L_c statistics test (Nyblom, 1989; Hansen, 1992) has also been employed to make the Granger causality tests more reliable. Table 5 shows the results of the above tests.

The *Sup-F*, *Mean-F*, and *Exp-F* tests show that BTC, PCI, EPU, and the VAR system have structural changes at a 1% significance level. These tests also show that the parameters presented in the variables would gradually change over time, at a 1% significance level, while the same happens at a 5% significance level for VAR. Additionally, the L_c statistics test demonstrates that the VAR system parameters cannot follow the random walk process at a level of 1%, which further proves that the parameters are marked by non-constancy. Furthermore, based on the stability tests, we have also found that the parameters exhibit unstable

Table 4
Full-Sample Granger Causality Test.

Tests	H ₀ : PCI does not Granger cause BTC Statistics	p-values	H ₀ : BTC does not Granger cause PCI Statistics	p-values
Bootstrap LR Test	1.486	0.690	1.956	0.570
Tests	H ₀ : EPU does not Granger cause BTC Statistics	p-values	H ₀ : BTC does not Granger cause EPU Statistics	p-values
Bootstrap LR Test	1.616	0.400	1.957	0.330

Table 5
Parameter Stability Test.

	BTC		PCI		VAR (3) System	
	Statistics	Bootstrap p-value	Statistics	Bootstrap p-value	Statistics	Bootstrap p-value
<i>Sup-F</i>	99.44***	<0.01	628.35***	<0.01	130.59***	<0.01
<i>Mean-F</i>	19.55***	<0.01	67.25***	<0.01	18.85**	<0.05
<i>Exp-F</i>	45.25***	<0.01	309.71***	<0.01	60.83***	<0.01
L_b^c	1.56**	<0.05	2.44***	<0.01	3.99***	<0.01
	BTC		GPU		VAR (3) System	
	Statistics	Bootstrap p-value	Statistics	Bootstrap p-value	Statistics	Bootstrap p-value
<i>Sup-F</i>	107.07***	<0.01	48.49***	<0.01	115.95***	<0.01
<i>Mean-F</i>	13.33***	<0.01	21.14***	<0.01	17.21**	<0.05
<i>Exp-F</i>	49.07***	<0.01	20.36***	<0.01	53.51***	<0.01
L_b^c	1.42*	<0.10	3.47***	<0.01	4.88***	<0.01

Notes: This study calculates p-values using 10,000 bootstrap repetitions.

*** and ** denote the significance at 10, 5 and 1 percent, respectively.

properties, due to the structural changes that have been experienced. Thus, the result of the full sample's causality analysis is inaccurate and biased. Thus, in order to identify the time-variation's interrelationship that might exist between these variables, we then conducted the bootstrap sub-sample rolling-window causality test. In particular, a time period spanning over 24-months⁴ has been taken into consideration as the width of the rolling window, which has the capability to make the results of the causal analysis more accurate.

According to the p-value presented in Fig. 3 (column 1), we can thus identify the degree and direction of the impact of PCI on BTC. It is observed that PCI does Granger cause BTC in these periods (2016:M05–2017:M03, 2018:M05–2018:M08, and 2020:M01–2020:M10), at a significance level of 10%. Moreover, the positive effects (2016:M05–2017:M03 and 2020:M01–2020:M10) and negative impacts (2018:M05–2018:M08) also exist from PCI to BTC. Following the same context, Fig. 5 (column 2) highlights that the null hypothesis which propagates that BTC does not have Granger cause PCI, is therefore rejected at a 10% significance level, during several of the sub-sample periods, including (2016:M10–2017:M03, 2018:M08–2019:M09, and 2019:M07–2019:M12). That is to say that, the results during these sub-periods, and the direction of the influence of BTC on PCI, are both positive and negative. Moreover, these results are also robust with the quantile-on-quantile, and the quantile based Granger causality methods.

The values shown in Fig. 6 (column 1) shed light on the revelation that, the null hypothesis that pertains to the EPU not having Granger cause BTC, is rejected at a 10% significance level, during the sub-sample periods that included (2015:M01–2015:M10, 2016:M02–2016:M05, 2018:M10–2019:M02, and 2020:M02–2020:M04). According to the results, during the span of these sub-periods, the impact of EPU on BTC, happens to be both positive and negative. Moreover, according to the p-values presented in Fig. 6 (column 2), we are also able to fathom the extent of the degree, and direction of the impact of BTC on EPU. It is evident that BTC does Granger cause in the periods that are marked by (2014:M11–2015:M05, 2016:M10–2016:M12, and 2018:M09–2018:M11), at a significance level of 10%. As a result, we can ascertain that there are negative and positive impacts of BTC on EPU, particularly during these periods of time. Moreover, these results are robust with the quantile-on-quantile, and the quantile based Granger causality methods.

5. Conclusion and policy implications

It is a widely believed notion that Bitcoin promotes global economic growth, by providing easier access to the capital and financial services.

⁴ Besides this, we changed the width to 20-, 28-, and 32- months, respectively, and conducted the robustness tests accordingly. Therefore, these results are similar to the empirical analysis that proposes that the rolling-window width is 24.

Bitcoin enables and has enabled in the past, many investors, including individuals, businesses, and governments, to grow and prosper. Unfortunately, the broad range of instabilities (political) and uncertainties (economic) can and have influenced the Bitcoin market, time and again. In this regard, this study evaluates how the political and economic uncertainties in the US, particularly during and after the US elections, have an impact on the Bitcoin price. In this paper, we also discuss how the Bitcoin market is mutually affected by the political uncertainty (PCI) and economic uncertainty (EPU) in the US, and whether Bitcoin has acted as a safe haven asset, particularly during the times of high uncertainties, and has emerged as a winner after the US factionalism. For the purpose of analyzing the relationship between BTC, PCI and EPU, we have primarily relied on the Wavelet-based quantile-on-quantile approach, and the quantile-based Granger causality. Also, in order to ensure the robustness of our results, we have also employed the bootstrap full sample and bootstrap rolling window causality tests, so as to analyze the impact of PCI and EPU on BTC. The empirical findings show that both PCI and EPU have negative and positive influences on BTC. The positive impact of PCI and EPU, on BTC, shows that Bitcoin can be regarded as a safe haven asset, and also a hedging instrument during the times when there is a high level of political and economic uncertainty in the country. However, as we have observed, there are also times when the impact of PCI and EPU on BTC is negative. This negative impact refutes Bitcoin's hypothesis as a safe haven asset, and can in fact be a warning sign for Bitcoin investors, who consider it as an effective hedge against the uncertainties. Moreover, these findings do not comply with the ICAPM, which indicates that the PCI and EPU have a positive influence on BTC. If we look at the matter in the other way around, BTC has a negative effect on PCI and EPU. This essentially suggests that the Bitcoin market should be taken into account, particularly when assessing US's political situation. By investigating the time and frequency domain decomposition, based on the relationship between BTC, PCI, and EPU, we have also been able to show that Bitcoin is not always a safe haven asset, as the relationship varies from the short to the long-run as well.

Therefore, understanding Bitcoin's hedging potential, with reference to US political climate, and with the uncertainty in economic environment, can teach some interesting lessons to both investors and regulators. Since the relationship varies and runs in cycles, it is important first to identify which cycle the investors are in. If investors are able to identify that they are in the cycle when uncertainties affect BTC positively, they can effectively use it as a hedging asset, and optimize their portfolio's performance. During these times, the fluctuations in PCI and EPU can thus be used by authorities to forecast the price of BTC, which could ultimately help to avoid any unnecessary volatility in the market. However, if the investors find themselves in the cycle where BTC is already on a decline, due to the Bitcoin bubble explosion, they must take an immediate exit from this market, and avoid any blind speculation, in

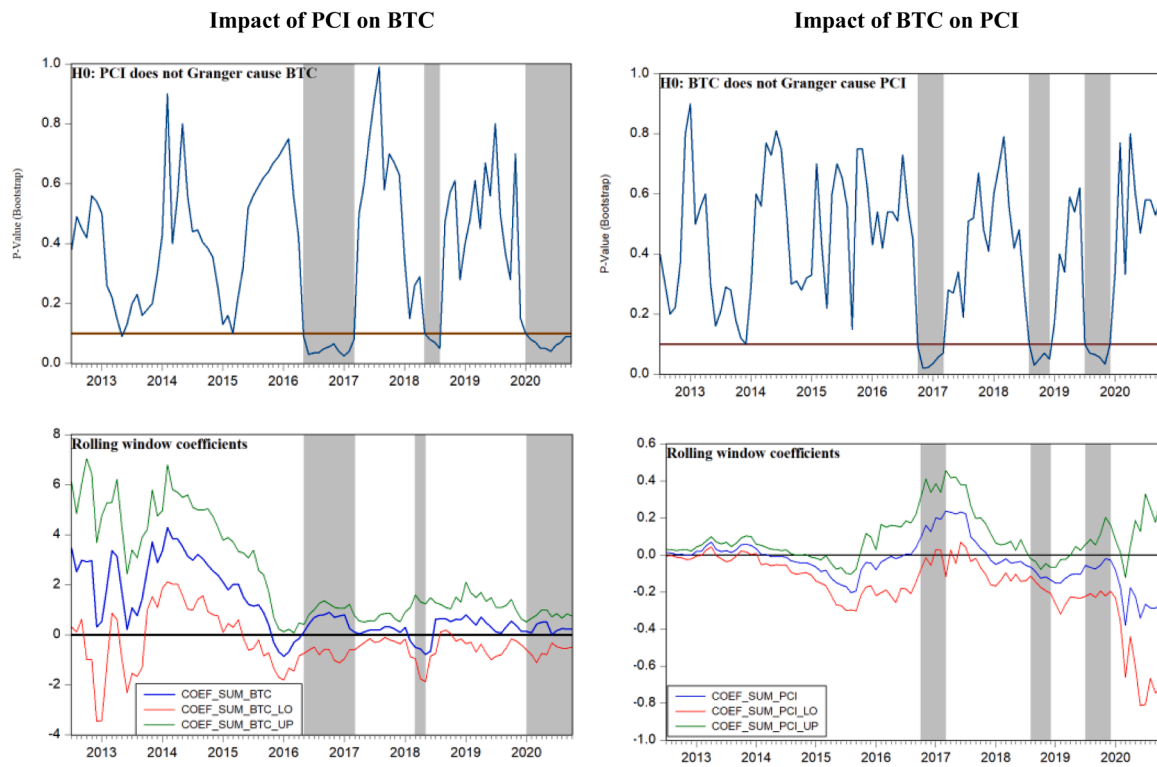


Fig. 5. Sub-sample bootstrap rolling-window Granger causality.

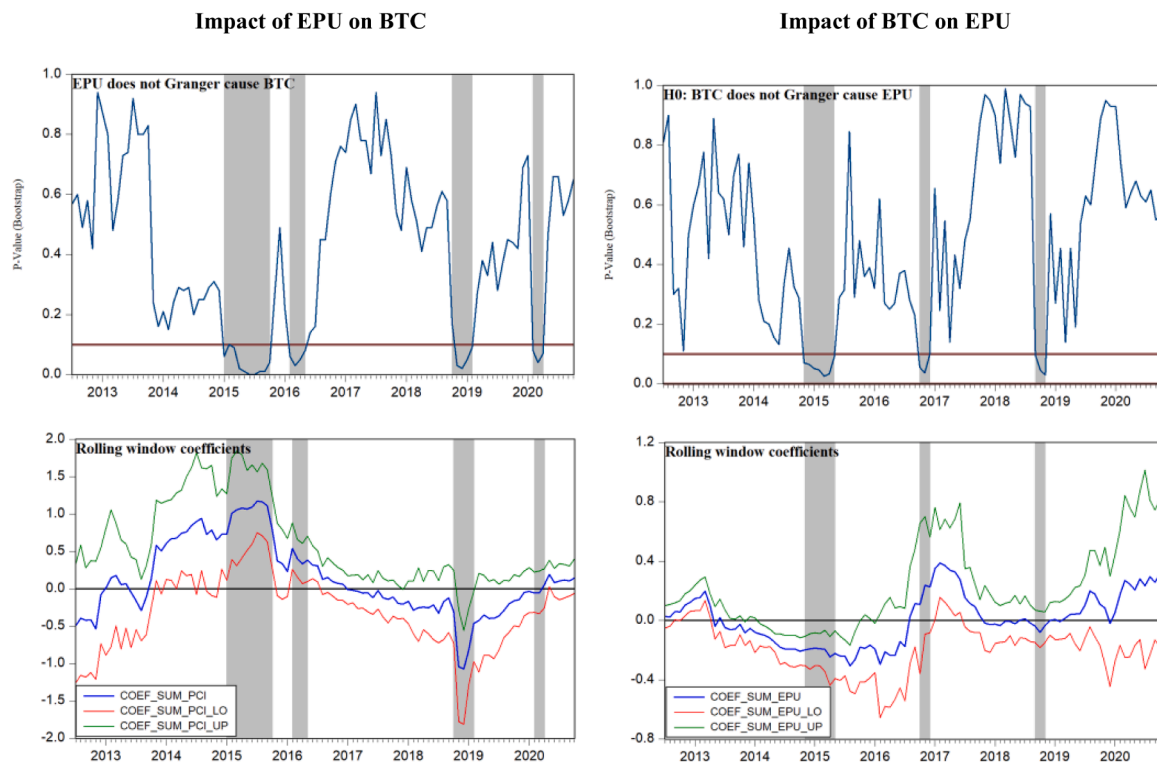


Fig. 6. Sub-sample bootstrap rolling-window Granger causality.

order to minimize the loss of investment and preserve their wealth. Given the heightened importance of Bitcoin in today's day and age, the latest encryption technology, network safety, and supervision should be improved and enhanced, so that more investors and traders can be attracted towards the Bitcoin market, which would ultimately help

stabilize the market, and also ensure its upward growth. Moreover, it is also important to note that since the last many periods, BTC has had some effects on PCI and EPU. This essentially highlights that the Bitcoin market can potentially be an important factor in understanding the uncertainties in the US politics and economy.

CRediT authorship contribution statement

Muhammad Umar: Conceptualization, Software, Data curation, Formal analysis, Supervision, Writing – review & editing. **Chi-Wei Su:** Software, Data curation, Supervision, Validation, Project administration, Writing – original draft. **Syed Kumail Abbas Rizvi:** Methodology, Visualization, Investigation, Project administration, Writing – review & editing. **Xue-Feng Shao:** Conceptualization, Validation, Project administration, Writing – review & editing.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.techfore.2021.120680](https://doi.org/10.1016/j.techfore.2021.120680).

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